Capital Adequacy

Lecture 12

BANK3011

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Objectives of this week's lecture

- Functions of capital.
- Development of the Basel Agreements.
- Definition of capital in Basel III.
- Basel III capital requirements.
- Calculation of credit risk-adjusted capital ratios using the BIS Standardized Approach.
- Criticisms against risk-based capital ratios.

1. Functions of capital

- What is capital? Definitions vary significantly across different perspectives:
  - Economist's perspective: an FI's capital, or owner's equity is the difference between the market values of its assets and liabilities, also known as net worth. This concept aligns with market value accounting.
  - Accounting perspective: Accountants typically focus on book value, which is the historical cost of assets minus liabilities as recorded in financial statements.
  - Regulatory perspective: Regulatory bodies prioritize financial stability. Regulatory capital requirements often rely on historical or book value accounting concepts and include various tiers (e.g., Tier 1 and Tier 2) and is used to assess capital adequacy.

1. Functions of capital

- To absorb operating and unanticipated losses with enough margin to inspire confidence and enable the FI to continue as a going concern.
- To protect uninsured depositors, bondholders, and creditors in the event of insolvency, and liquidation.
- To protect the deposit insurance funds and the taxpayers
- To fund the real investments to provide better financial services

1. Functions of capital – Example of bank capital absorbing losses

- Consider the following market value balance sheet of an FI.

Table 1: Market-value-based balance sheet before loan losses

Assets ($m)	Amount	Liabilities ($m)	Amount
Securities	70	Deposits	85
Loans	30	Net worth	15
Total assets	100	Total liabilities + equity	100

- The FI is solvent on a market value basis.

1. Functions of capital – Example of bank capital absorbing losses

- The marking-to-market method allows balance sheet values to reflect current rather than historic prices.
- Consider a fall in the market value of loans to $10 (a fall of $20m).

Table 2: Market-value-based balance sheet after loan losses

Assets ($m)	Amount	Liabilities ($m)	Amount
Securities	70	Deposits	85
Loans	**10**	Net worth	**-5**
Total assets	**80**	Total liabilities + equity	**80**

- The FI is no more solvent on a market value basis. its net worth has declined from $15 to -$5. The owners' net worth stake has been completely wiped out.
- After the liquidation of the remaining $80 in assets, depositors would get only 80/85 in dollars, without deposit insurance. The FI's capital is used to absorb (partially) the losses.
- The example also shows that market valuation of the balance sheet produces an economically accurate picture of the net worth and thus the solvency position of an FI.

1. Functions of capital – Example of bank capital absorbing losses

- However, the FI's balance sheet based on book value could remain unchanged.

Table 3: Book-value-based balance sheet after loan losses

Assets ($m)	Amount	Liabilities ($m)	Amount
Securities	70	Deposits	85
Loans	30	Net worth	15
Total assets	100	Total liabilities + equity	100

- Note that Table 3 looks identical to Table 1.
- With book value accounting, FIs have discretion in how and when they report loan losses on their balance sheets.
- This flexibility allows them to strategically manage the recognition of these losses and their subsequent effect on capital.

1. Functions of capital – Example of bank capital absorbing losses

- For example, the FI could just record an increase in loan loss provisions to reflect their expected loan losses (e.g., $5m).

Table 4: Book-value-based balance sheet with loan loss provisions

Assets ($m)	Amount	Liabilities ($m)	Amount
Securities	70	Deposits	85
Loans	30	Net worth	10
less loan loss provisions	(5)		
Total assets	95	Total liabilities + equity	95

1. Functions of capital – Market value vs. book value

- Obviously, market-value-based view of capital allows for a more accurate and comprehensive description of FIs' financial health.
  - If regulators close a FI before it's market value of capital reaches zero, liability holders will not lose.
- But not all assets and liabilities are valued as fair value (market value).
  - All trading assets, marketable securities ("available for sale") are marked to market.
  - Loans and debt securities held for investment or to maturity are carried at amortized cost (book value).

1. Functions of capital – Market value vs. book value

- Why not market value for all?
  - ➢ Difficult to implement, especially for small banks, building societies and credit unions with large amounts of non-traded assets
  - ➢ Introduces unnecessary variability into an FI's earnings
  - ➢ FIs are less willing to take long-term asset exposures such as commercial mortgages and business loans, since long-term assets are more interest rate sensitive.

2. Basel agreement or accord

- Actual capital ratios applied can be country-specific, determined by national regulators (e.g. APRA in Australia). However, the Basel Accords provide the global framework for these capital ratios.
- The Basel Committee on Banking Supervision (BCBS) was formed in 1974 by banking authorities from the G-10 nations to “strengthen the soundness and stability of the international banking system while maintaining sufficient consistency that capital adequacy regulation will not be a significant source of competitive inequality among internationally active banks.”
  - ✓ It was designed to level the playing field for FIs globally

1. Functions of capital

- Capital Ratio

Q1: How is capital defined?



$$Capital\ Ratio = \frac{Capital}{Some\ measure\ of\ Assets}$$



Q2: How are assets measured? Especially, how are assets adjusted for credit risk, market risk and operational risk? (For teaching simplicity, we will only consider credit risk).

2. Basel agreement or accord – Basel I

- Two capital ratios
  - ➢ Tier 1 capital ratio
    - ✓ Primarily composed of common equity, retained earnings, and disclosed reserves, less goodwill and other intangibles.
    - ✓ Calculation: Tier 1 Capital/Risk-Weighted Assets (RWAs)
    - ✓ Minimum: 4%
  - ➢ Total capital ratio: minimum 8%
- Features
  - ➢ Basel I introduced the systematic measurement of capital adequacy through the use of risk-weighted assets.
  - ➢ Basel I utilized RWA to account for the varying risk levels of different asset classes, including both on-balance-sheet and off-balance-sheet exposures.
  - ➢ Credit Risk Focus: Basel I primarily recognized credit risk in the calculation of risk-weighted assets. It did not initially incorporate market risk or operational risk

2. Basel agreement or accord – Basel II

Basel II comprised three pillars:

1. minimum capital requirements, which sought to develop and expand the standardised rules set out in the 1988 Accord

2. supervisory review of an institution's capital adequacy and internal assessment process

3. effective use of disclosure as a lever to strengthen market discipline and encourage sound banking practices

- The measurement of capital did not change markedly in Basel II.
- The measurement of risk was significantly enhanced to include operational risk, some market risks in the banking book, and risks associated with securitisation.

Under Basel II, two options are allowed for banks to measure their credit risk:

1. Standardized approach. Similar to Basel I, but more risk-sensitive.

2. Internal ratings-based (IRB) approach. Banks can use their internal rating system or credit scoring models to assess their portfolios, subject to regulatory approval.

Three options are available for measuring operational risk

2. Basel agreement or accord

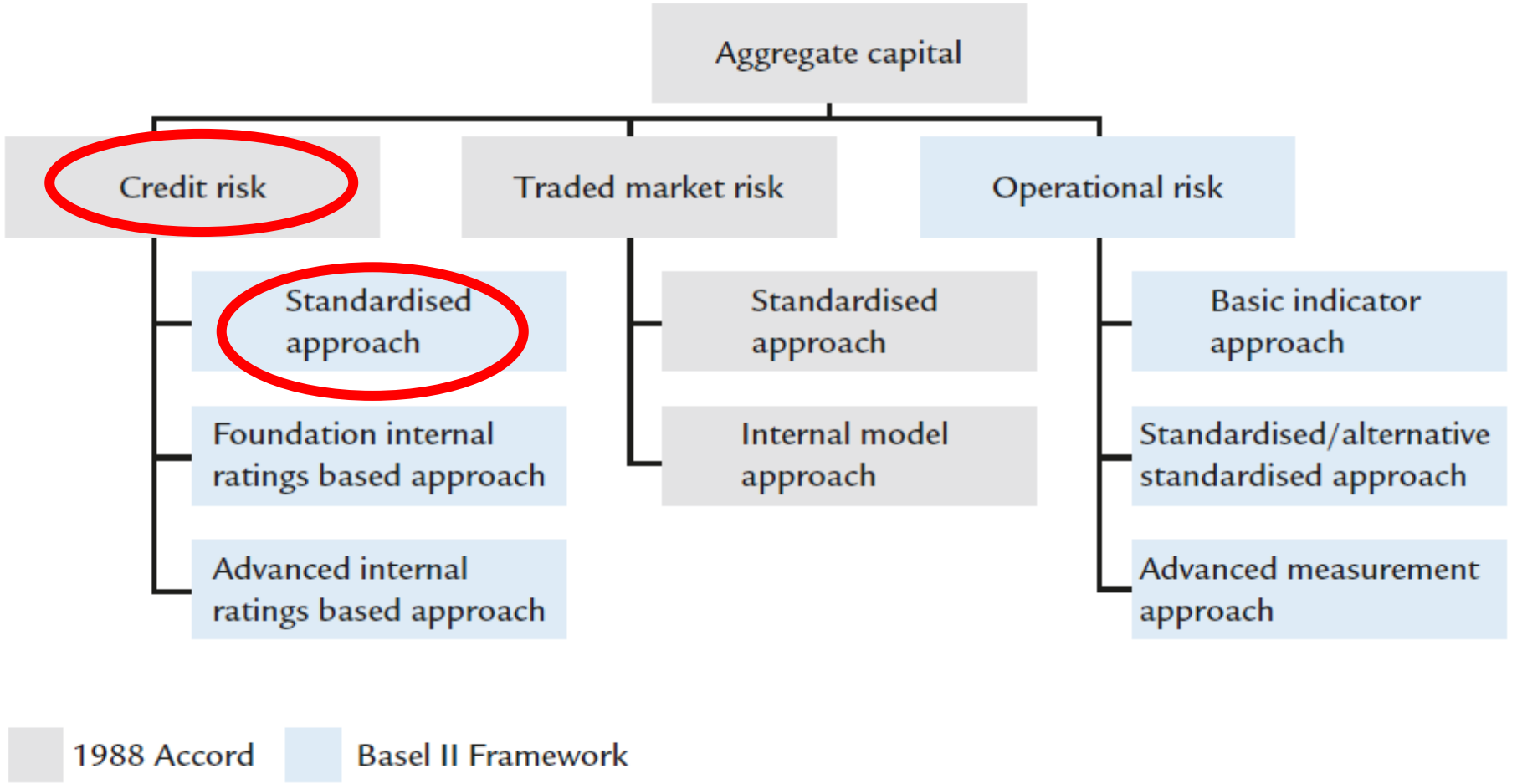
- Basel II and III Pillars of Capital Regulation
  - ✓ Basel II consisted of three mutually reinforcing pillars, which together contribute to the safety and soundness of the financial system.
    - ➢ Pillar 1 covered regulatory minimum capital requirements for credit, market, and operational risk.
    - ➢ Pillar 2 stressed the importance of the regulatory supervisory review process as a critical complement to minimum capital requirements. Basel II created a process through which regulators ensure that each DI has sound internal processes in place to assess the adequacy of its capital and set targets for capital that are commensurate with the DI's specific risk profile and control environment.

2. Basel agreement or accord

- Basel II and III Pillars of Capital Regulation
  - ✓ Basel II consisted of three mutually reinforcing pillars, which together contribute to the safety and soundness of the financial system.
    - ➢ In Pillar 3, the BIS encourages market discipline by developing a set of requirements on the disclosure of capital structure, risk exposures, and capital adequacy. Such disclosure requirements allow market participants to assess critical information describing the profile and capital adequacy of DIs.

2. Basel agreement or accord

- Basel II Framework of Calculating Minimum Capital



2. Basel agreement or accord

- Weakness of Basel II Revealed By the Financial Crisis
  - ✓ The financial crisis of 2008-09 revealed weaknesses with Basel II.
    - ➢ Not enough focus on liquidity – may have had adequate capital, but not enough liquid assets
    - ➢ Market risk models significantly underestimated market risk during GFC

2. Basel agreement or accord – Basel III

Basel III (passed in 2010, phased in 2013-2019)

- Basel III is broader in perspective than just a revision of capital, capital adequacy, risk measurement and supervision. It introduced **macroprudential** measures, targeting the protection of the whole financial system.
- Three pillars similar to in Basel II, but with significant enhancements.

3. Basel III Improvements

- 2010/2011 – Basel III framework finalised
  - ✓ Improved quality and quantity of capital, especially Tier 1 (refine T1 into common equity tier 1 CET1 and additional tier 1 AT1)
  - ✓ Introduced a higher liquidity standard (LCR and NSFR)
  - ✓ Introduced an internationally harmonized (non-risk based) leverage ratio to serve as a backstop to the risk-based capital requirements

3. Basel III Improvements

- 2010/2011 – Basel III framework finalised
  - ✓ Introduced a capital conservation buffer
    - ➢ 2½ per cent of RWAs will provide banks with additional capital that they can draw upon in stressed periods. This buffer is entirely in the form of common equity tier 1 capital (CET1). As such, the minimum CET1 capital requirement plus the buffer will be 7 per cent of RWAs, once both are fully phased in. If a bank's CET1 ratio falls below 7 per cent, constraints on its capital distributions will be imposed (as well as other supervisory measures).

3. Basel III Improvements

- 2010/2011 – Basel III framework finalised
  - ✓ Introduced countercyclical buffer
    - ➢ 2½ per cent of RWAs (entirely in the form of CET1) may be imposed by the relevant national authority during periods when system-wide risk is building up (credit growth)
  - ✓ additional capital surcharge for systemically important institutions (SIFIs)

3. Basel III Improvements

- Pillar 2 calls for enhanced incentives for banks to better manage risk and returns over the long term, more stress testing, and implementation of sound compensation practices.
  - ✓ For example, compensation should be sensitive to risk outcomes over a multi-year horizon. This is typically achieved through arrangements that defer compensation until risk outcomes have been realised, and may include so-called “clawback” provisions whereby compensation is reduced or reversed if employees generate exposures that cause the bank to perform poorly in subsequent years or if the employee has failed to comply with internal policies or legal requirements.
- Pillar 3 calls for enhanced disclosure of risks, such as those relating to securitization exposures and sponsorship of off-balance-sheet vehicles.

3. Basel III Improvements

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Basel II:

Pillar 1	Pillar 2	Pillar 3
Calculation of regulatory minimum capital requirements	Regulatory supervisory review so as to complement and enforce minimum capital requirements calculated under Pillar 1	Requirements on rules for disclosure of capital structure, risk exposures, and capital adequacy so as to increase FI transparency and enhance market/investor discipline
1. Credit risk: on-balance-sheet and off-balance-sheet (Standardized vs. Internal Ratings–Based Approach)		
2. Market risk (Standardized vs. Internal Ratings–Based Approach)		
3. Operational risk (Basic Indicator vs. Standardized vs. Advanced Measurement Approach)		

Basel III:

Enhanced minimum capital and liquidity requirements	Enhanced supervisory review for firmwide risk management and capital planning	Enhanced risk disclosure and market discipline
1. Liquidity risk		

4. Basel III Ratios

- Tier 1 Leverage Ratio
  - ✓ A simple, non-risk based leverage ratio of Tier 1 capital to total exposure (on- and off-balance-sheet) to serve as a backstop to the risk-based capital ratios.
    - ➢ Initially set at a 3% minimum. Final adjustment in 2017.

4. Basel III Ratios

- Tier 1 Leverage Ratio
  - ✓ Example: Risk-based capital ratio is usually higher than the simple leverage ratio when many assets receive less than 100% risk weights.

Assets		Risk Weights	Liabilities + Equity	
T-Bills	$1m	0%	Deposits	$ 1.9 m
Commercial loans	$1m	100%	Common equity	$ 100,000

- $Tier\ 1\ capital\ ratio = \frac{\$100{,}000}{\$1m*0\%+\$1m*100\%} = 10\%$

- $Tier\ 1\ leverage\ ratio = \frac{\$100{,}000}{\$1m+\$1m} = 5\%$

4. Basel III Ratios

- DIs must calculate and monitor key ratios:

$$\text{Common Equity Tier 1 Capital ratio} = \frac{\text{Common Equity Tier 1 Capital}}{\text{Total risk-weighted assets}}$$

$$\text{Tier 1 Capital ratio} = \frac{\text{Tier 1 Capital}}{\text{Total risk-weighted assets}}$$

$$\text{Total Capital ratio} = \frac{\text{Total Capital}}{\text{Total risk-weighted assets}}$$

4. Basel III Ratios

- Credit Risk-Adjusted On-Balance-Sheet Assets
  - ✓ Risk weights
    - ➢ Classified into one of five risk categories (0-5).
    - ➢ Categories 0-1 have the lowest possible risk and are assigned a risk weight of 0%, while category 5 is assigned a risk weight of 150%.

4. Basel III Ratios

Risk Categories

Category 1 (0% weight)

Cash, Federal Reserve Bank balances, securities of the U.S. Treasury, OECD governments, some U.S. agencies, and loans to sovereigns with an S&P credit rating of AA− or better.

Category 2 (20% weight)

Cash items in the process of collection. U.S. and OECD interbank deposits and guaranteed claims.

Some non-OECD bank and government deposits and securities. General obligation municipal bonds.

Some mortgage-backed securities. Claims collateralized by the U.S. Treasury and some other government securities.

Loans to sovereigns with an S&P credit rating of A+ to A− . Loans to banks and corporates with an S&P credit rating of AA− or better.

Category 3 (50% weight)

Loans fully secured by first liens on one- to four-family residential properties. Other (revenue) municipal bonds. Loans to sovereigns with an S&P credit rating of BBB+ to BBB−. Loans to banks and corporates with an S&P credit rating of A+ to A−.

4. Basel III Ratios

Risk Categories
Category 4 (100% weight)
Loans to sovereigns with an S&P credit rating of BB+ to B−. Loans to banks with a credit rating of BBB+ to B−. Loans to corporates with a credit rating of BBB+ to BB− and unrated C&I loans.
All other on-balance-sheet assets not listed above, including loans to private entities and individuals, some claims on non-OECD governments and banks, real assets, and investments in subsidiaries.
Category 5 (150% weight)
Loans to sovereigns, banks, and securities firms with an S&P credit rating below B−. Loans to corporates with a credit rating below BB−.

4. Basel III Ratios

- Credit Risk-Adjusted Off-Balance-Sheet Assets
  - ✓ A DI's off-balance-sheet (OBS) activities represent contingent rather than actual claims against the DI.
  - ✓ Regulations require that capital be held not against the full face value of these items, but an amount equivalent to any eventual on-balance-sheet credit risk these securities might create for a DI.
  - ✓ To calculate the credit risk-adjusted asset values for these OBS items, we first convert them into credit equivalent amounts – amounts equivalent to an on-balance-sheet item.
  - ✓ We then assign the OBS credit equivalent amount to an appropriate risk category to calculate the credit risk-adjusted value for these items.

4. Basel III Ratio

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TABLE 20–9 Bank's Balance Sheet under Basel III (in millions of dollars)

Weight	Assets		Liabilities/Equity		Capital Class
0%	Cash	$ 8	Demand deposits	$ 150	
	Balances due from Fed	13	MMDAs deposits	500	
	Treasury bills	60	CDs	380	
	Long-term Treasury securities	50	Fed funds purchased	80	
	Long-term government agencies (GNMAs)	42			
20	Items in process of collection	10	Convertible bonds	10	Tier II
	Long-term government agencies (FNMAs)	10	Subordinated bonds	10	Tier II
	Munis (general obligation)	20			
	Loans to countries with OECD CRC rating of 2	55			
	Loans to foreign banks in country with OECD CRC rating of 2	10	Perpetual preferred stock (nonqualifying)	5	Tier II
50	University dorm bonds (revenue)	34			
	Residential 1–4 family mortgages	308	Retained earnings	40	CET 1
	Category 1, loan-to-value ratio between 60% and 80%				
	Loans to foreign banks in country with OECD CRC rating of 3	75	Common stock	30	CET 1
100	Commercial loans	390			
	Third world loans, B+ rated	108	Noncumulative perpetual preferred stock (qualifying)	10	Additional Tier I
	Premises, equipment	22			
150	Loans to countries with OECD CRC rating of 7	10		$1,215	
N/A	Reserve for loan losses	(10)			Tier II
	Total assets	$1,215			

Off-Balance-Sheet Items:

$80 m in two-year loan commitments to a U.S. corporation
$10 m direct credit substitute standby letters of credit issued to a U.S. corporation
$50 m in commercial letters of credit issued to a U.S. corporation

One fixed–floating interest rate swap for four years with notional dollar value of $100 m and replacement cost of $3 m
One 2-year Euro$ contract for $40 m with a replacement cost of −$1 m

4. Basel III Ratios

- Example: Calculating Credit RWA
  - ✓ Credit risk-adjusted on-balance-sheet assets = 0*(8 +13+60+50+42) + 0.2*(10+10+20+55+10) + 0.5*(34+308+75) +1*(390+108+22) +1.5*(10) = $764.5 million
  - ✓ Credit risk-adjusted asset value of OBS guaranty contracts

Item	Face Value	Conversion factor	Credit Equivalent Amount	Risk Weight	RWA
2-year loan commitment	$80 m	0.5	$40 m	1.0	$40 m
SLC	$10 m	1.0	$10 m	1.0	$10 m
Commercial LC	$50 m	0.2	$10 m	1.0	$10 m

- Example: Calculating Credit RWA

✓ Credit equivalent amount of OBS derivative contracts

Item	Notional amount	Conversion factor	Potential exposure	Current exposure	Credit Equivalent Amount
4-year fixed-floating interest rate swap	$100 m	0.5%	$0.5 m	$3 m	$3.5 m
2-year forward foreign exchange contract	$40 m	5%	$2 m	$0 m	$2 m

✓ Credit risk-adjusted asset value of OBS derivative contracts

= $5.5 m *1.0 =$5.5 million

✓ Total Credit RWA = $764.5 + $60 + $5.5 = $830 million

4. Basel III Ratios

- Example: Calculating Capital Ratios
  - ✓ CET1 = $40 m + $30 m = $70 million
  - ✓ Tier 1 capital = $70 m + $10 m= $80 million
  - ✓ Tier 2 capital = $10 m (convertible debt) + $10 m (subordinated debt) +
  - ✓ $5 m (Non-qualifying perpetual preferred stock) + $10 m (Loan Loss Reserve) = $35 million
    - ➢ $CET\ 1\ capital\ ratio$= ($70 m)/($830 m)=8.43%
    - ➢ $Tier\ 1\ capital\ ratio$= ($70 m+$10 m)/($830 m)=9.64%
    - ➢ $Total\ capital\ ratio$= ($70 m+$10 m+$35 m)/($830 m)=13.68%
    - ➢ $Leverage\ ratio$= ($70 m+$10 m)/($1,215 m)=6.58%

4. Basel III Ratios

6. National Bank has the following balance sheet (in millions) and has no off-balance-sheet activities. (Textbook Chap 20, Q18)

Assets		**Liabilities and Equity**	
Cash	$20	Deposits	$960
Treasury bills	40	Subordinated debentures	25
Residential mortgages		Common stock	45
(category 1; loan-to-value		Retained earnings	40
ratio = 70%; risk-weight =50%)	600	Total liabilities and equity	$1,090
Business loans (risk-weight=100%)	430		
Total assets	$1,090		

4. Basel III Ratios

a. What is the CET1 risk-based ratio?

Risk-adjusted assets = $20x0.0 + $40x0.0 + $600x0.5 + $430x1.0 = $300 + $430= $730.

The CET1 risk-based ratio is ($45 + $40)/$730 = 0.1164 or 11.64 percent.

NOTE: Cash and T-bills are deemed to be risk-free assets.

b. What is the Tier I risk-based capital ratio?

Risk-adjusted assets = $20x0.0 + $40x0.0 + $600x0.5 + $430x1.0 = $730.

Tier I capital ratio = ($45 + $40)/$730 = 0.1164 or 11.64 percent.

Note: As there is no additional tier 1 capital in this case, the answer is the same as for CET1 above.

c. What is the total risk-based capital ratio?

The total risk-based capital ratio = ($45 + $40 + $25)/$730 = 0.1507 or 15.07 percent.

NOTE: The total risk-based capital ratio includes tier 1 and tier 2 capital.

d. What is the leverage ratio?

The leverage ratio is ($45 + $40)/$1,090 = 0.078 or 7.80 percent.

NOTE: The leverage ratio is not risk-based.

4. Basel III Ratios

- Criticisms of the Risk-Based Capital Ratios
  - ✓ Validity of risk-weights
  - ✓ Risk weights based on credit ratings
  - ✓ It largely ignores credit risk portfolio diversification opportunities or concentration risks
  - ✓ Basel III will greatly raise the cost of regulation by adding new levels of complexity.
  - ✓ Australia is becoming the first OECD country to phase out Additional Tier 1 (AT1) hybrid bonds from bank capital frameworks (government brief)

Summary

- Capital is required within FIs to absorb operating and unexpected losses with enough margin to inspire confidence and to enable the FI to continue as a going concern.
- Also serves as a source of financing to purchase and invest in assets.
- The Basel Accord has undergone major changes especially in response to the 2008-2009 GFC
- Minimum capital requirements are now higher but remain risk-based